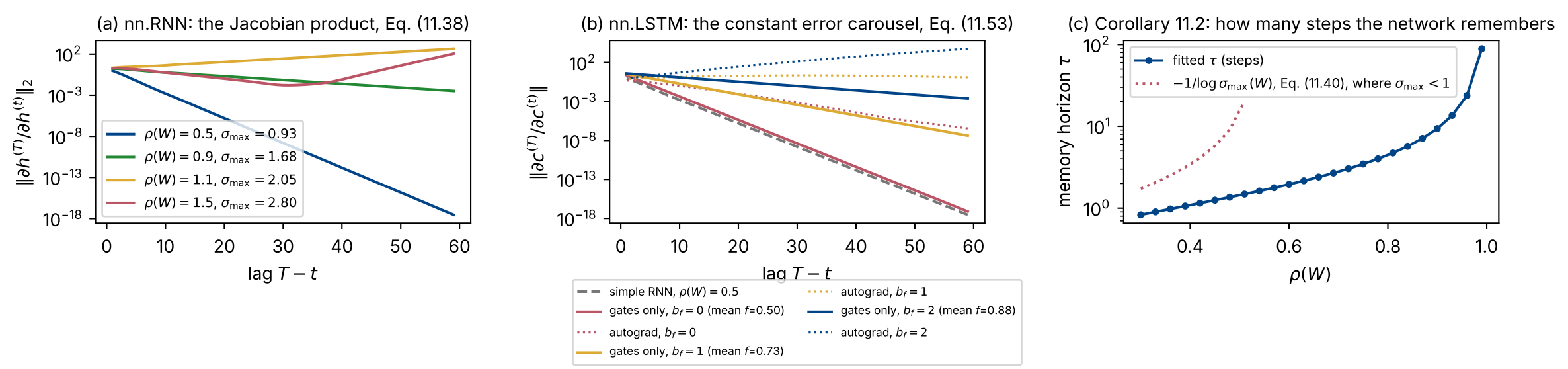
(a) nn.RNN: the Jacobian product, Eq. (11.38)
$\|\partial h^{(T)}/\partial h^{(t)}\|_2$
$\rho(W) = 0.5$, $\sigma_{\max} = 0.93$
$\rho(W) = 0.9$, $\sigma_{\max} = 1.68$
$\rho(W) = 1.1$, $\sigma_{\max} = 2.05$
$\rho(W) = 1.5$, $\sigma_{\max} = 2.80$
lag $T - t$
(b) nn.LSTM: the constant error carousel, Eq. (11.53)
$\|\partial c^{(T)}/\partial c^{(t)}\|$
lag $T - t$
simple RNN, $\rho(W) = 0.5$
gates only, $b_f = 0$ (mean f=0.50)
autograd, $b_f = 0$
gates only, $b_f = 1$ (mean f=0.73)
autograd, $b_f = 1$
gates only, $b_f = 2$ (mean f=0.88)
autograd, $b_f = 2$
(c) Corollary 11.2: how many steps the network remembers
memory horizon τ
fitted τ (steps)
$-1/\log \sigma_{\max}(W)$, Eq. (11.40), where $\sigma_{\max} < 1$
$\rho(W)$